SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I _U.Ananya(192424382)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:
School of Computing → /25
School of Electronics → /26
School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:
Software Development → **/24**
Quality Assurance → **/25**
Human Resources → **/26**
Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin	Shows good understandi ng with minor gaps	Partial understandin g; misses key aspects	Misinterpret s or fails to understand the scenario

		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganize d and difficult to understand

Scenario-Based Questions

1. Smart Campus Network Subnet Allocation

Given data

A Smart Campus has been allocated the IP address block 192.168.100.0/24. The network must be divided among three departments.

- School of Computing → /25
- School of Electronics → /26
- School of Mechanical Engineering → /27

Step 1: Calculate subnet masks and host capacity

The formulas used are:

$$Total\ addresses = 2^{32-prefixlength}$$

$$Usablehosts = Totaladdresses - 2$$

Department	**CIDR prefix**	**Subnet mask**	**Total addresses**	**Usable hosts**
Computing	/25	255.255.255.128	128	126
Electronics	/26	255.255.255.192	64	62
Mechanical Engineering	/27	255.255.255.224	32	30

Step 2: Allocate the subnet addresses

School of Computing – /25

A /25 subnet contains 128 addresses. Starting from 192.168.100.0, the subnet covers addresses from .0 to .127.

- Network address: 192.168.100.0
- First valid host: 192.168.100.1
- Last valid host: 192.168.100.126
- Broadcast address: 192.168.100.127
- Usable hosts: 126

School of Electronics – /26

The next available address is 192.168.100.128. A /26 subnet contains 64 addresses.

- Network address: 192.168.100.128
- First valid host: 192.168.100.129
- Last valid host: 192.168.100.190
- Broadcast address: 192.168.100.191
- Usable hosts: 62

School of Mechanical Engineering – /27

The next available address is 192.168.100.192. A /27 subnet contains 32 addresses.

- Network address: 192.168.100.192
- First valid host: 192.168.100.193
- Last valid host: 192.168.100.222
- Broadcast address: 192.168.100.223
- Usable hosts: 30

Final subnet allocation table

Details	**Computing**	**Electronics**	**Mechanical Engineering**

Network address	192.168.100.0	192.168.100.128	192.168.100.192
Valid host range	.1 – .126	.129 – .190	.193 – .222
Broadcast address	.127	.191	.223
Usable hosts	126	62	30

Step 3: Calculate the remaining unused range

The last allocated address is 192.168.100.223.

Therefore, the remaining unused range is:

192.168.100.224 – 192.168.100.255

Total unused addresses:

$256 - 224 = 32$

Conclusion: The Smart Campus network is divided into three non-overlapping subnets. Computing supports 126 hosts, Electronics supports 62 hosts, and Mechanical Engineering supports 30 hosts. The remaining 32 addresses can be reserved for future use.

2. Multinational Software Company Network Allocation

Given data

The company has been allocated 10.10.0.0/16.

Department requirements:

- Software Development → /24
- Quality Assurance → /25

- Human Resources → /26
- Executive Management → /27

For this solution, the subnets are allocated consecutively from the beginning of the /16 block, without overlap.

Step 1: Calculate the subnet masks and host capacities

Department	**Prefix**	**Subnet mask**	**Total addresses**	**Usable hosts**
Software Development	/24	255.255.255.0	256	254
Quality Assurance	/25	255.255.255.128	128	126
Human Resources	/26	255.255.255.192	64	62
Executive Management	/27	255.255.255.224	32	30

Step 2: Allocate Software Development

A /24 subnet contains 256 addresses.

- Network address: 10.10.0.0/24
- First valid host: 10.10.0.1
- Last valid host: 10.10.0.254
- Broadcast address: 10.10.0.255
- Usable hosts: 254

Step 3: Allocate Quality Assurance

The next available address is 10.10.1.0. A /25 subnet contains 128 addresses.

- Network address: 10.10.1.0/25

- First valid host: 10.10.1.1
- Last valid host: 10.10.1.126
- Broadcast address: 10.10.1.127
- Usable hosts: 126

Step 4: Allocate Human Resources

The next available address is 10.10.1.128. A /26 subnet contains 64 addresses.

- Network address: 10.10.1.128/26
- First valid host: 10.10.1.129
- Last valid host: 10.10.1.190
- Broadcast address: 10.10.1.191
- Usable hosts: 62

Step 5: Allocate Executive Management

The next available address is 10.10.1.192. A /27 subnet contains 32 addresses.

- Network address: 10.10.1.192/27
- First valid host: 10.10.1.193
- Last valid host: 10.10.1.222
- Broadcast address: 10.10.1.223
- Usable hosts: 30

Final subnet allocation table

Department	**Network address**	**Valid host range**	**Broadcast address**	**Usable hosts**
Software Development	10.10.0.0/24	10.10.0.1 – 10.10.0.254	10.10.0.255	254

Quality Assurance	10.10.1.0/25	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	10.10.1.128/26	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	10.10.1.192/27	10.10.1.193 – 10.10.1.222	10.10.1.223	30

Step 6: Determine the unused IP address range

The original /16 block covers 10.10.0.0 through 10.10.255.255.

The last allocated address is 10.10.1.223.

Therefore, the remaining unused range is:

10.10.1.224 – 10.10.255.255

Total allocated addresses:

$256 + 128 + 64 + 32 = 480$

Total addresses in the original block:

$2^{32-16} = 65,536$

Remaining addresses:

$65,536 - 480 = 65,056$

Conclusion: Each department receives a subnet suited to its requirements. A total of 480 addresses are allocated, leaving 65,056 addresses for future expansion.

3. IPv6 Migration in a Smart Hospital

Given data

The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

IPv6 addresses contain 128 bits represented by eight hexadecimal groups. Each group contains 16 bits.

Step 1: Compress the IPv6 address

Two main rules are used for IPv6 compression:

1. Leading zeros in each hexadecimal group can be removed.
2. A consecutive sequence of zero groups can be replaced with ::, and this replacement may be used only once.

Original address:

2001:0db8:abcd:0000:0000:0000:0000:0000

After removing leading zeros:

2001:db8:abcd:0:0:0:0:0

Replace the consecutive zero groups with ::.

Compressed address:

2001:db8:abcd::

Step 2: Expand the compressed address

The compressed address is:

2001:db8:abcd::

It contains three visible groups. Therefore, the :: represents five zero groups to make eight groups in total.

Expanded address:

2001:0db8:abcd:0000:0000:0000:0000:0000

This is identical to the original IPv6 address.

Step 3: Identify the network prefix

The given prefix length is /64, which means that the first 64 bits represent the network portion.

Since each group contains 16 bits:

$$64 \div 16 = 4 groups$$

The first four groups are:

2001:0db8:abcd:0000

Therefore, the network prefix is:

2001:db8:abcd:0::/64

The remaining 64 bits are available for interface identifiers.

Step 4: Calculate the total number of host addresses

IPv6 address length = 128 bits.

Network bits = 64 bits.

Remaining bits:

$$128 - 64 = 64$$

Number of possible interface identifier values:

$$2^{64} = 18,446,744,073,709,551,616$$

Thus, the /64 subnet has 2^{64} possible interface identifier values.

IPv6 does not use broadcast addresses like IPv4. The value above represents the size of the address space; actual host assignment may follow additional IPv6 conventions.

Conclusion: The compressed IPv6 address is 2001:db8:abcd::, and its network prefix is 2001:db8:abcd:0::/64. It provides 64 bits for interface identifiers, supporting an extremely large number of connected medical devices and IoT sensors.

4. ISP Router and Routing Table Analysis

Given data

An Internet Service Provider connects multiple customer networks through a core router. The routing table contains the following entries:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP address:

192.168.2.45

Step 1: Identify the destination subnet

The destination address is 192.168.2.45.

A /24 subnet uses the subnet mask 255.255.255.0. The first three octets identify the network.

Therefore, the destination belongs to:

192.168.2.0/24

The address range of this subnet is:

- Network address: 192.168.2.0
- Valid host range: 192.168.2.1 – 192.168.2.254
- Broadcast address: 192.168.2.255

The destination address 192.168.2.45 is a valid host address in this subnet.

Step 2: Identify the matching routing table entry

The router compares the destination IP address with its routing table.

Routing table entry	Match?	Reason
192.168.1.0/24	No	Different third octet
192.168.2.0/24	Yes	Matches the destination network
192.168.3.0/24	No	Different third octet

Therefore, the matching routing table entry is:

192.168.2.0/24

Step 3: Determine the next-hop network or interface

After identifying the matching route, the router forwards the packet through the associated outgoing interface or next-hop router.

For example, the routing table might contain:

Destination network	Next hop / outgoing interface
192.168.1.0/24	Interface A
192.168.2.0/24	Interface B
192.168.3.0/24	Interface C

In this example, the packet for 192.168.2.45 would be forwarded through Interface B.

However, the actual next-hop IP address or interface cannot be determined from the given destination prefixes alone. The real routing table must provide that information. If the network is directly connected, the router forwards the packet on the connected interface and resolves the destination's link-layer address as needed.

Step 4: Determine the default route

A default route is used when no more specific matching route exists.

In IPv4, the default route is represented as:

0.0.0.0/0

It may point to an upstream ISP router, for example:

0.0.0.0/0 → configured upstream next hop

If the destination does not match any specific route, the router uses this default route if one is configured. If no matching route or usable default route exists, the packet is normally dropped.

Final result

Task	**Answer**
Destination subnet	192.168.2.0/24
Matching route	192.168.2.0/24
Destination IP	192.168.2.45
Next-hop interface	Must be obtained from the router's actual routing table

Default route	0.0.0.0/0

Conclusion: The destination 192.168.2.45 belongs to 192.168.2.0/24. The router selects the matching route and forwards the packet using its configured next hop or outgoing interface. If no specific route matches, it can use the default route 0.0.0.0/0 when available.

5. Communication Between Administration and Finance LANs

Given data

A company has two departments connected through a Layer-3 router.

- Administration LAN: 192.168.10.0/26
- Finance LAN: 192.168.10.64/26

Router gateway addresses:

- Administration Gateway: 192.168.10.1
- Finance Gateway: 192.168.10.65

Step 1: Calculate the subnet mask and block size

Both LANs use the /26 prefix.

Subnet mask:

255.255.255.192

Number of host bits:

$32 - 26 = 6$

Total addresses:

$2^6 = 64$

Usable hosts:

$64 - 2 = 62$

Block size:

$256 - 192 = 64$

Therefore, each subnet contains 64 addresses and supports 62 usable host addresses.

Step 2: Determine the Administration LAN details

The Administration LAN is 192.168.10.0/26.

Its address range is 192.168.10.0 – 192.168.10.63.

- Network address: 192.168.10.0
- First valid host: 192.168.10.1
- Last valid host: 192.168.10.62
- Broadcast address: 192.168.10.63
- Default gateway: 192.168.10.1
- Usable hosts: 62

The gateway 192.168.10.1 is a valid host address and is assigned to the router interface connected to Administration.

Step 3: Assign IP addresses to three Administration computers

Three suitable host addresses are:

Device	**IP address**	**Subnet mask**	**Default gateway**
Administration PC 1	192.168.10.2	255.255.255.192	192.168.10.1
Administration PC 2	192.168.10.3	255.255.255.192	192.168.10.1

Administration PC 3	192.168.10.4	255.255.255.192	192.168.10.1

These addresses are within the valid host range and do not conflict with the router's gateway address, assuming they are not already assigned.

Step 4: Determine the Finance LAN details

The Finance LAN is 192.168.10.64/26.

Its address range is 192.168.10.64 – 192.168.10.127.

- Network address: 192.168.10.64
- First valid host: 192.168.10.65
- Last valid host: 192.168.10.126
- Broadcast address: 192.168.10.127
- Default gateway: 192.168.10.65
- Usable hosts: 62

The gateway 192.168.10.65 is a valid host address and is assigned to the router interface connected to Finance.

Step 5: Assign IP addresses to three Finance computers

Three suitable host addresses are:

Device	**IP address**	**Subnet mask**	**Default gateway**
Finance PC 1	192.168.10.66	255.255.255.192	192.168.10.65
Finance PC 2	192.168.10.67	255.255.255.192	192.168.10.65

Finance PC 3	192.168.10.68	255.255.255.192	192.168.10.65

These addresses are within the Finance LAN's valid host range and do not overlap with the network, broadcast, or gateway addresses.

Step 6: Explain communication between the LANs

Administration and Finance belong to different subnets.

- Administration devices use 192.168.10.1 as their default gateway.
- Finance devices use 192.168.10.65 as their default gateway.
- When a computer needs to communicate with a device in the other subnet, it sends the packet to its default gateway.
- The Layer-3 router examines the destination IP address and forwards the packet through the interface connected to the destination subnet.

This communication requires appropriate router configuration and any necessary access-control rules.

Final subnet allocation table

Details	**Administration LAN**	**Finance LAN**
Network address	192.168.10.0/26	192.168.10.64/26
Subnet mask	255.255.255.192	255.255.255.192
Valid host range	192.168.10.1 – 192.168.10.62	192.168.10.65 – 192.168.10.126
Broadcast address	192.168.10.63	192.168.10.127

Default gateway	192.168.10.1	192.168.10.65
Usable hosts	62	62

Conclusion: The two LANs have separate, non-overlapping /26 subnets. Each supports 62 usable hosts. The router provides communication between the departments, while the configured default gateways allow computers to send traffic to destinations outside their own subnet.